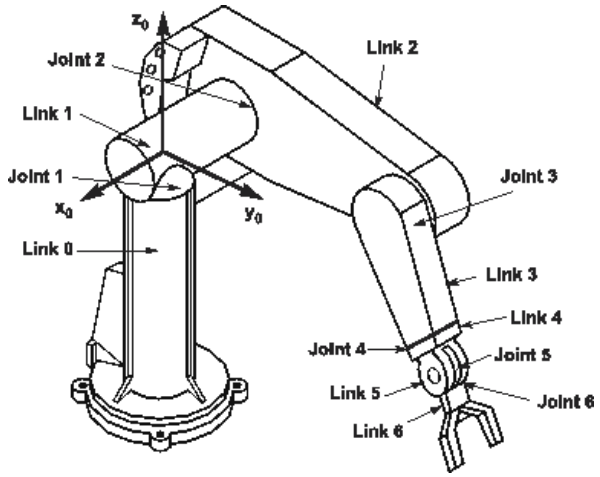
Lecture 3: Other Orientation Representations

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What did we learn last time?

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

What are we doing today?

1. Introduction to Robotics & Recap of Linear Algebra and Mathematical Notation
2. Translations & Rotation Matrices
3. **Other Representations for Orientation** ***(Today)***
4. Transformation Matrices
5. Forward Kinematics & DH Parameters
6. Schematic Representations of Robots, FK from DH, DH Parameters Practice
7. Inverse Kinematics
8. Kinematic Simulation
9. Angular Velocities, Velocity Propagation & Trajectory Generation
10. The Jacobian Matrix, Singularities
11. Analytical & Geometrical Jacobian, Numerical IK, Statics
12. Manipulability, More on the Robotic Systems Toolbox

Topics for Today

Part I:

- The **”problem”** with **orientation**
- **”Warm-up” exercise**/discussion
- **Overview** of orientation/rotation representations

Part II:

- **Angle-Set Conventions**
  - **Fixed angles**
  - **Euler angles**
- **Gimbal Lock**

Part III:

Equivalent **angle-axis** (Euler vector)

Part I: What is this whole fuss about orientation?

(or why we are using more than one lecture on it)

The Concept of Orientation

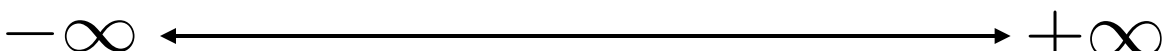
How would you represent an orientation:

- If we lived in a **1D world**?
- If we lived in a **2D world**?
- If we lived in a **3D world**?

Why is Orientation Difficult?

Position is **all nice and easy** for our brains.

- We like $\mathbb{R}$. It is easy to relate to:

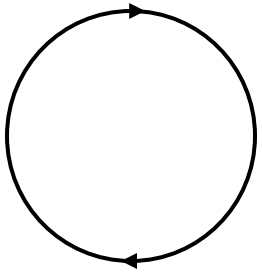


$$+\infty \neq -\infty$$



Orientation is a ***funky concept*** to your brain, in terms of mathematics.

- Orientation does not live in $\mathbb{R}$. It is related to circles and spheres and lives in $SO(3)$ (in 3D). Funny things happen:

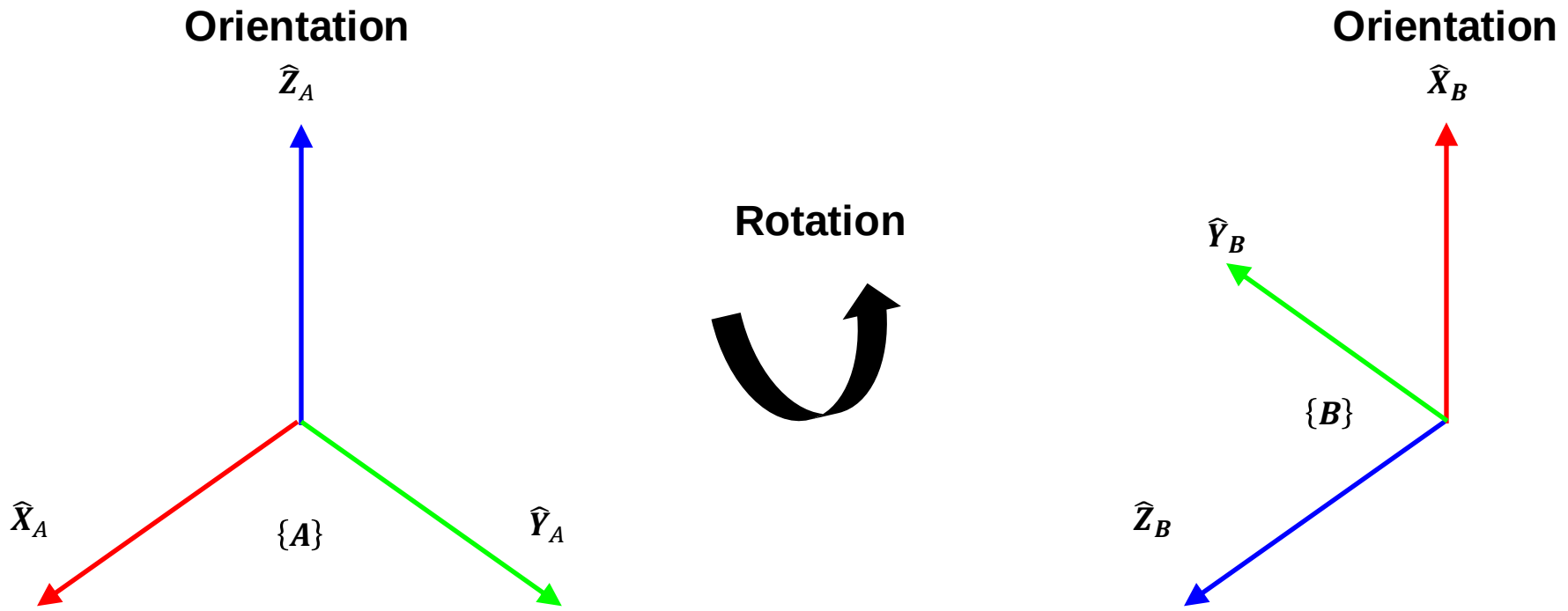




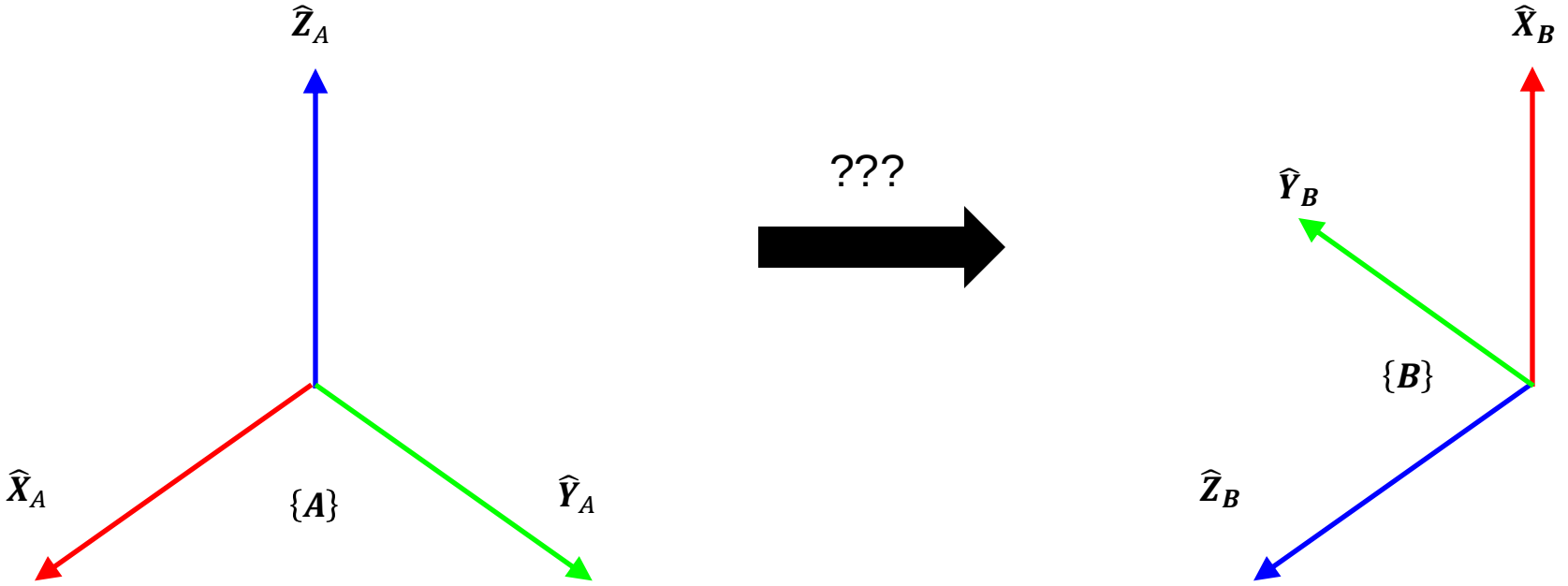
$$0 = 2\pi$$?!?!

Rotation vs. Orientation

- **Orientation describes a state**.
- **Rotation** is an **operation**, i.e. it describes how to get from one orientation to another.



How do you get from {A} to {B}?

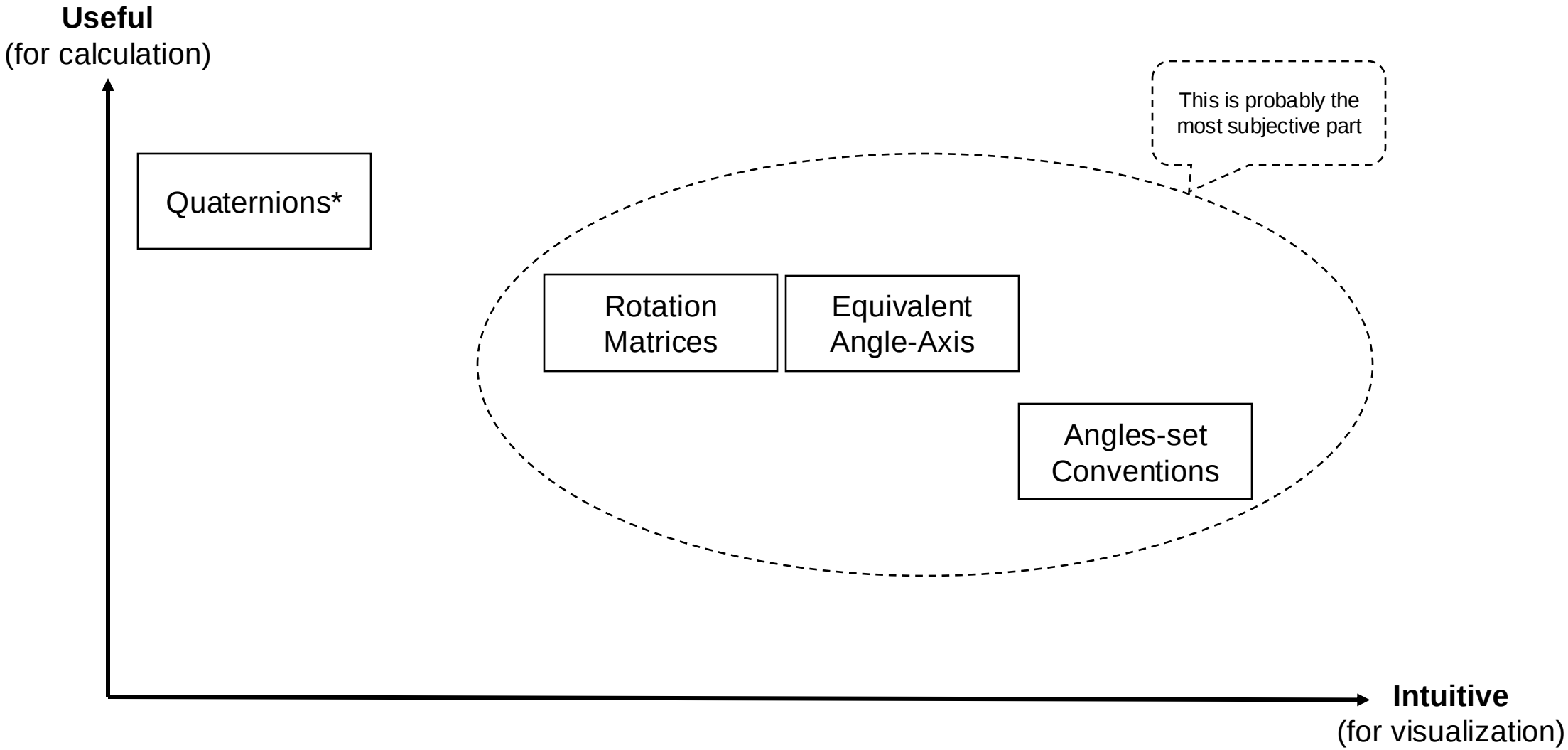


Splitting the **3D problem** into a **sequence of rotations** is one **solution***:

- You may think **this is easy**…
- …so why did I say orientation is difficult?
- Well, this **solution carries other problems**, as we will see.

Note: Called **Euler angles.*

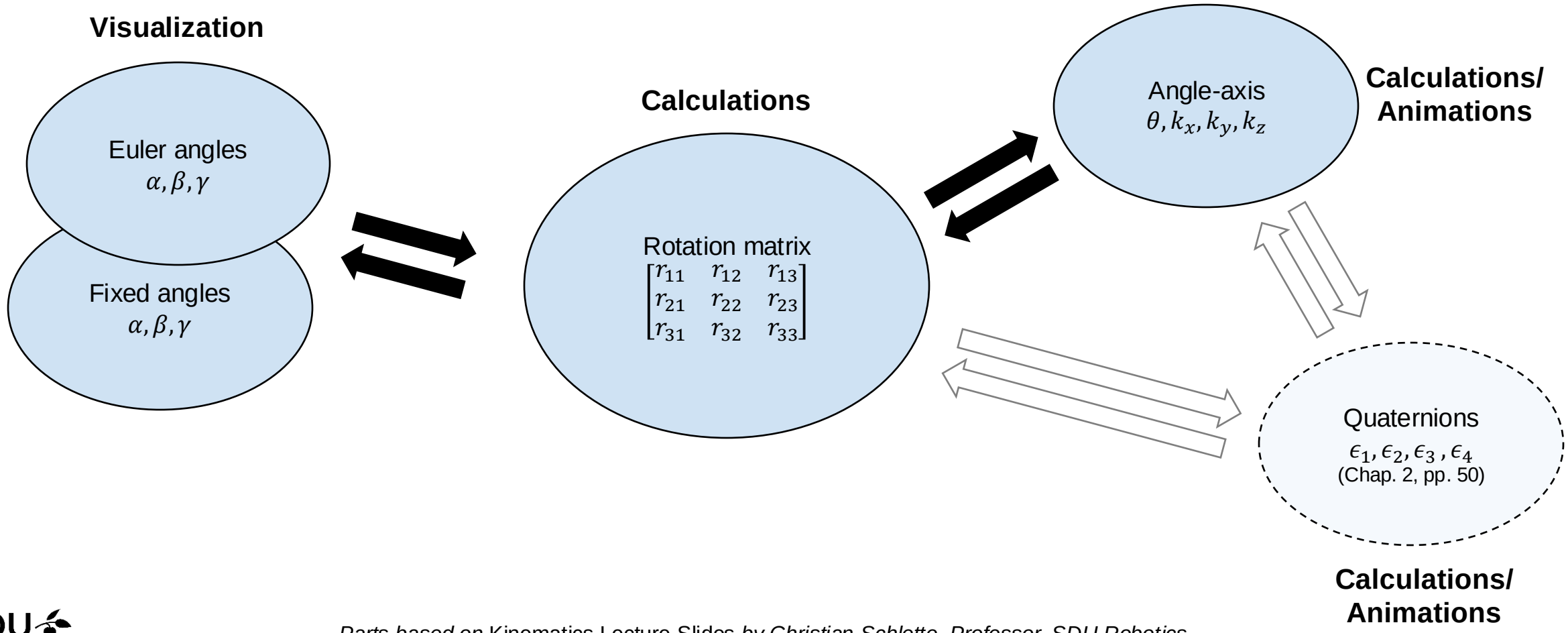
(Subjective) Overview of Orientation Representations



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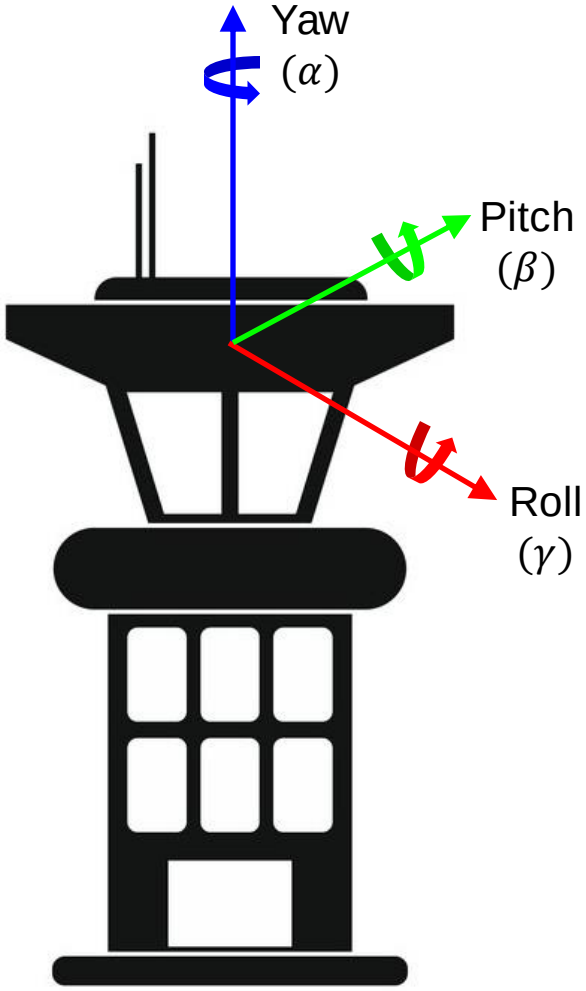
Quaternions are **wonderful and extremely powerful.*
*Also, they live in a **4-dimensional space** and will therefore make your brain explode if you think too much about them.*

Orientation/Rotation Representations: Use-Cases

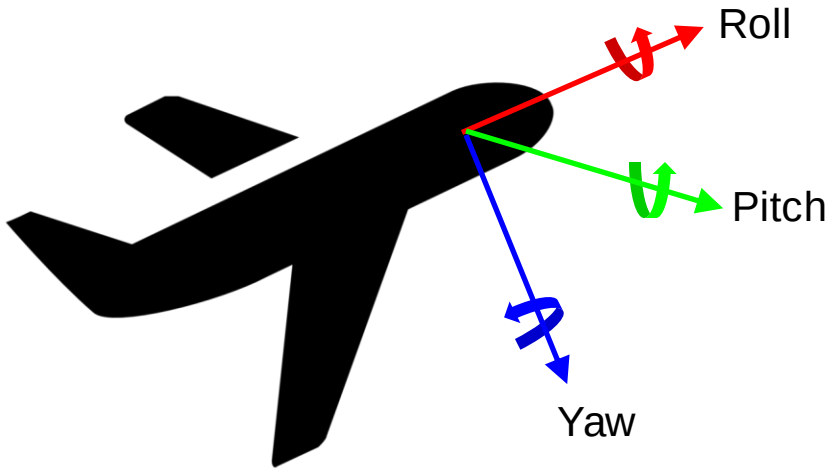


Part II: Angle-Set Conventions

Angle-Set Conventions



Fixed Angles



Euler Angles

Angle-set conventions represent orientation as **three separate rotations around different axes**.

- For **fixed angles**, the **coordinate system stays fixed/still**.
- For **Euler angles**, the **coordinate system rotates**.

The rotations can be applied in **many orders**.

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All 24 Angle-Set Conventions

12 Euler angles sets

$\boldsymbol{R}_{X'Y'Z'}$	$\boldsymbol{R}_{X'Z'Y'}$	$\boldsymbol{R}_{Y'X'Z'}$	$\boldsymbol{R}_{Y'Z'X'}$	$\boldsymbol{R}_{Z'X'Y'}$	$\boldsymbol{R}_{Z'Y'X'}$
$\boldsymbol{R}_{X'Y'X'}$	$\boldsymbol{R}_{X'Z'X'}$	$\boldsymbol{R}_{Y'X'Y'}$	$\boldsymbol{R}_{Y'Z'Y'}$	$\boldsymbol{R}_{Z'X'Z'}$	$\boldsymbol{R}_{Z'Y'Z'}$

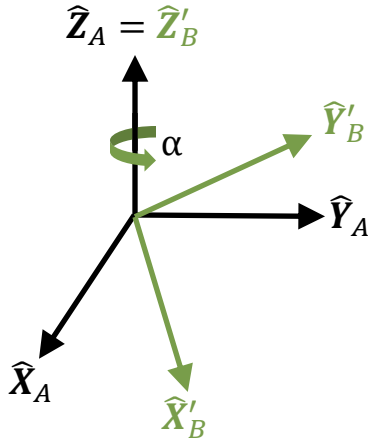
12 Fixed angle sets

$\boldsymbol{R}_{XYZ}$	$\boldsymbol{R}_{XZY}$	$\boldsymbol{R}_{YXZ}$	$\boldsymbol{R}_{YZX}$	$\boldsymbol{R}_{ZXY}$	$\boldsymbol{R}_{ZYX}$
$\boldsymbol{R}_{XYX}$	$\boldsymbol{R}_{XZX}$	$\boldsymbol{R}_{YXY}$	$\boldsymbol{R}_{YZY}$	$\boldsymbol{R}_{ZXZ}$	$\boldsymbol{R}_{YZY}$

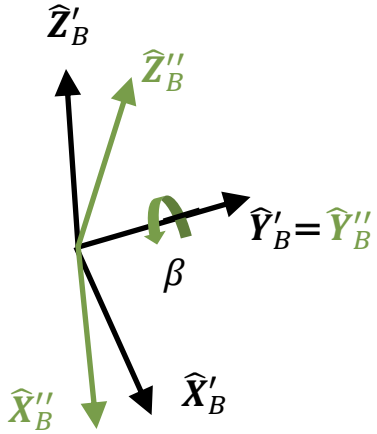
Euler Angles Z-Y-X

Euler angles: Each rotation is performed about an axis of the **coordinate system from the last step**.

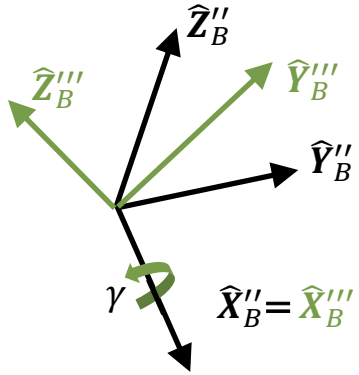
1. Start with a known frame $\{\boldsymbol{A}\}$. Rotate first about $\widehat{\boldsymbol{Z}}_A$ by angle α.



2. Then, rotate about the resulting $\widehat{\boldsymbol{Y}}'_B$ by angle β.

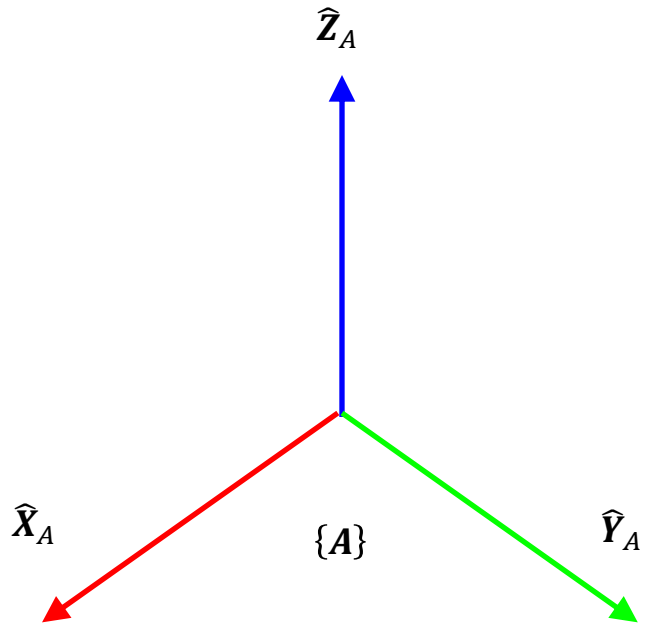


3. Finally, rotate about the resulting $\widehat{\boldsymbol{X}}''_B$ by angle γ.

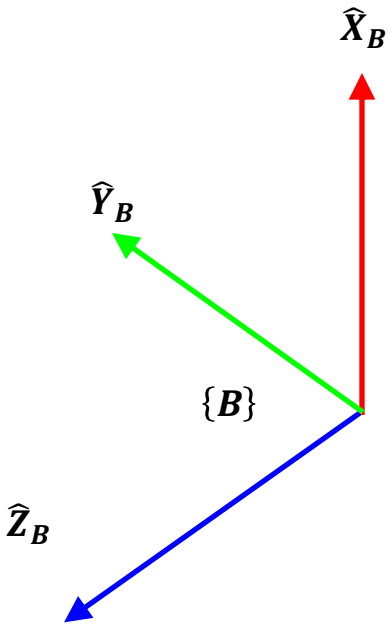


Euler Angles Z-Y-X to Rotation Matrix

You have **already done this**! Just **right-multiply three rotation matrices around each axis in order**:



$$Z = 90°$$
$$Y = 90°$$
$$X = 90°$$



$$^{A}_{B}\boldsymbol{R}_{Z'Y'X'}(\alpha,\beta,\gamma) = \boldsymbol{R}_Z(\alpha)\boldsymbol{R}_Y(\beta)\boldsymbol{R}_X(\gamma) = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

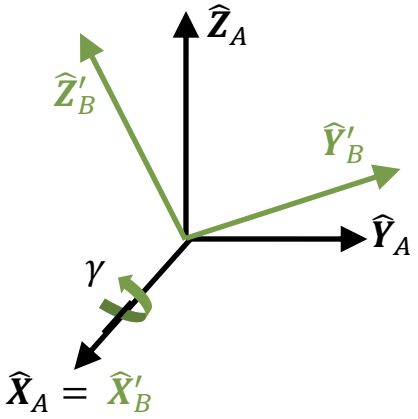
Stack from left to right
(i.e. multiply each new rotation on the right)

$$c\alpha = \cos\alpha\,, \qquad s\alpha = \sin\alpha$$

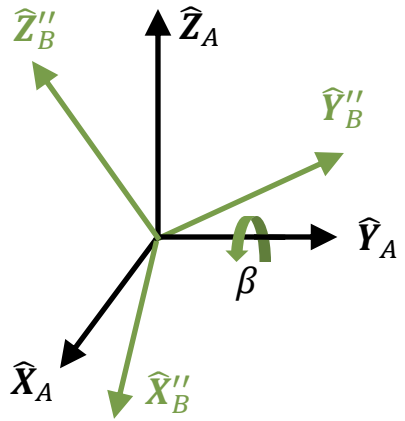
Fixed Angles X-Y-Z (Roll-Pitch-Yaw)

Fixed angles: Each rotation is performed about an axis of the same **fixed reference coordinate system**.

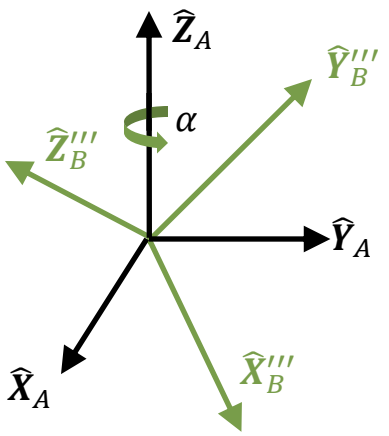
1. Start with a known frame $\{A\}$. Rotate first about $\hat{X}_A$ by angle γ.



2. Then, rotate about the $\hat{Y}_A$ by angle β.



3. Finally, rotate about the $\hat{Z}_A$ by angle α.



Fixed Angles X-Y-Z to Rotation Matrix

Opposite of Euler angles: **<u>left-multiply</u> three rotation matrices around each axis in order**:

$$ {}^{A}_{B}\boldsymbol{R}_{XYZ}(\gamma,\beta,\alpha) = \boldsymbol{R}_{Z}(\alpha)\boldsymbol{R}_{Y}(\beta)\boldsymbol{R}_{X}(\gamma) = \boldsymbol{???} $$

Stack from right to left
(i.e. multiply each new rotation on the left)

In MATLAB, try the following:

Using combinations of your three functions *rotx(), roty()*, *rotz():*

1. Calculate:
   a) The rotation matrix for the Euler Angles Z-Y-X, for $\alpha = 35°, \beta = 70°, \gamma = 20°$.
   b) The rotation matrix for the Fixed Angles X-Y-Z (RPY), for $\gamma = 20°, \beta = 70°, \alpha = 35°$.

2. Calculate:
   a) The rotation matrix for the Euler Angles Z-Y-Z, for $\alpha = 80°, \beta = 125°, \gamma = -70°$.
   b) The rotation matrix for the Fixed Angles Z-Y-Z, for $\gamma = -70°, \beta = 125°, \alpha = 80°$.

Notice anything interesting?

In MATLAB, try the following:

Using combinations of your three functions *rotx(), roty()*, *rotz():*

1. Calculate:
   a) The rotation matrix for the Euler Angles Z-Y-X $(35°, 70°, 20°)$.
   b) The rotation matrix for the Fixed Angles X-Y-Z (RPY) $(20°, 70°, 35°)$.

2. Calculate:
   a) The rotation matrix for the Euler Angles Z-Y-Z $(80°, 125°, -70°)$.
   b) The rotation matrix for the Fixed Angles Z-Y-Z $(-70°, 125°, 80°)$.

Notice anything interesting?

Relation between Fixed and Euler Angles

Fixed angles for a certain order are equal to Euler angles with the opposite order!

$$ {}^{A}_{B}\boldsymbol{R}_{XYZ}(\gamma,\beta,\alpha) = {}^{A}_{B}\boldsymbol{R}_{Z'Y'X'}(\alpha,\beta,\gamma) $$

$$ {}^{A}_{B}\boldsymbol{R}_{YZX}(\gamma,\beta,\alpha) = {}^{A}_{B}\boldsymbol{R}_{X'Z'Y'}(\alpha,\beta,\gamma) $$

$$ {}^{A}_{B}\boldsymbol{R}_{ZXY}(\gamma,\beta,\alpha) = {}^{A}_{B}\boldsymbol{R}_{Y'X'Z'}(\alpha,\beta,\gamma) $$

etc.

What is happening?

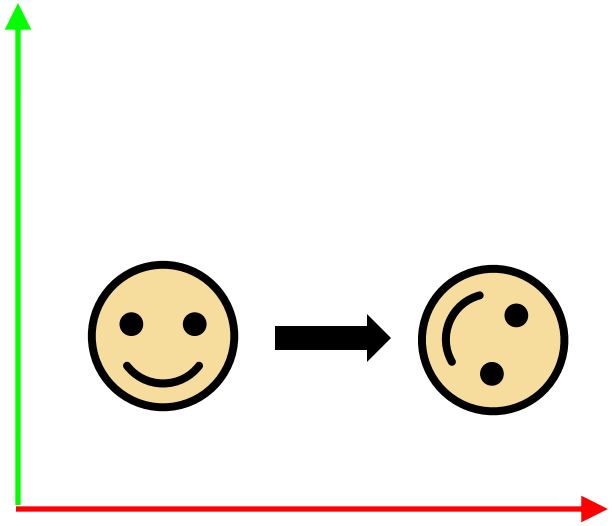
How can we interpret the two approaches geometrically?

Local vs. Global Coordinate Frames
(and Pre/Post-multiplication)

It comes down to: are we **moving** or **fixing** the reference frame?

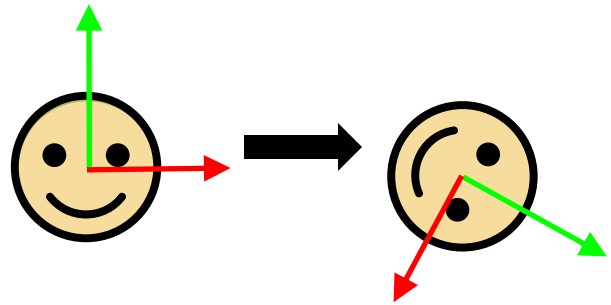
Fixed Angles

- **Global coordinate frame,** does not move
- **Left-multiplied**
- For a robot: the **base** is a **fixed frame**



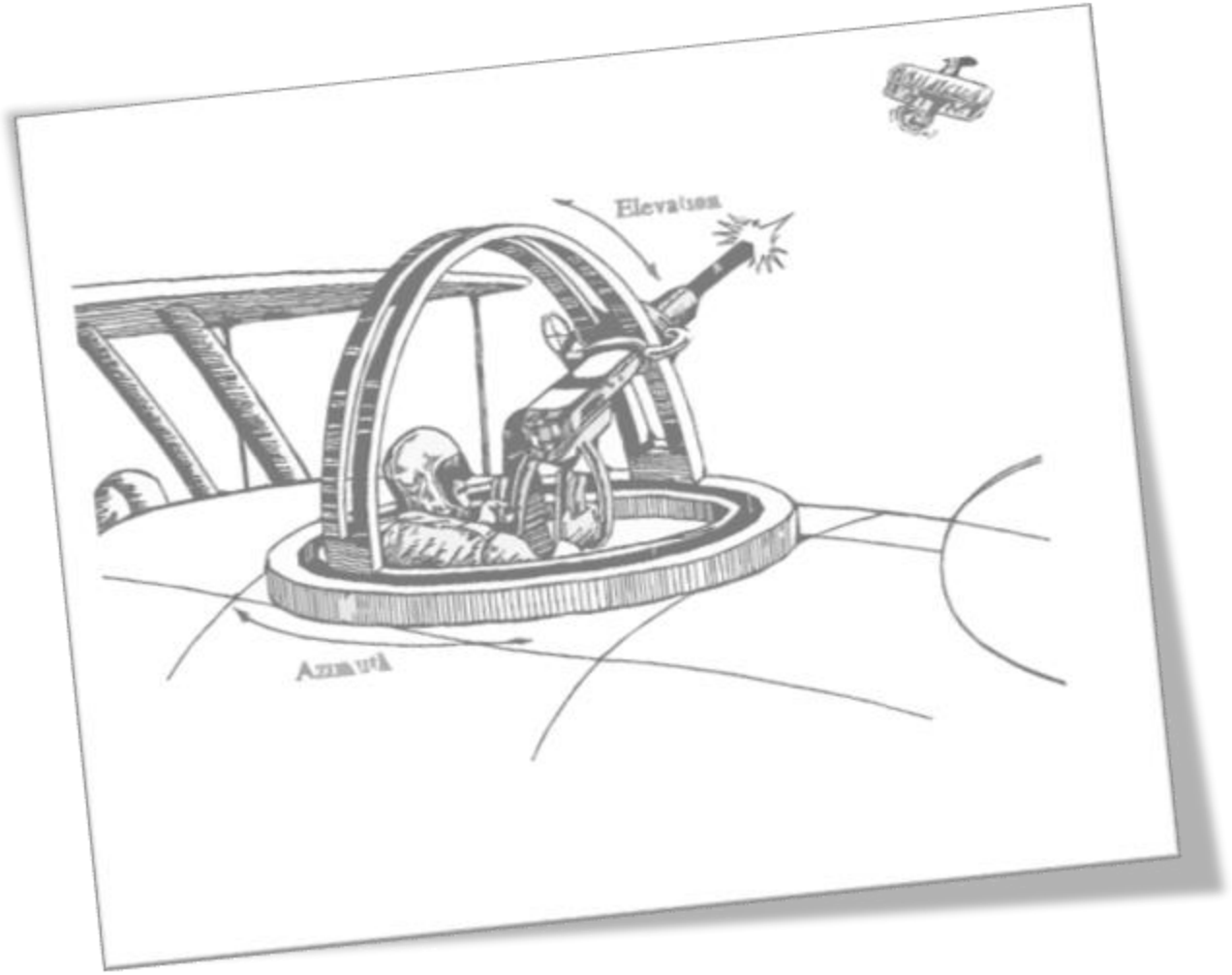
Euler Angles

- **Local coordinate frame,** moves
- **Right-multiplied**
- For a robot, the **tool** is a **moving frame**



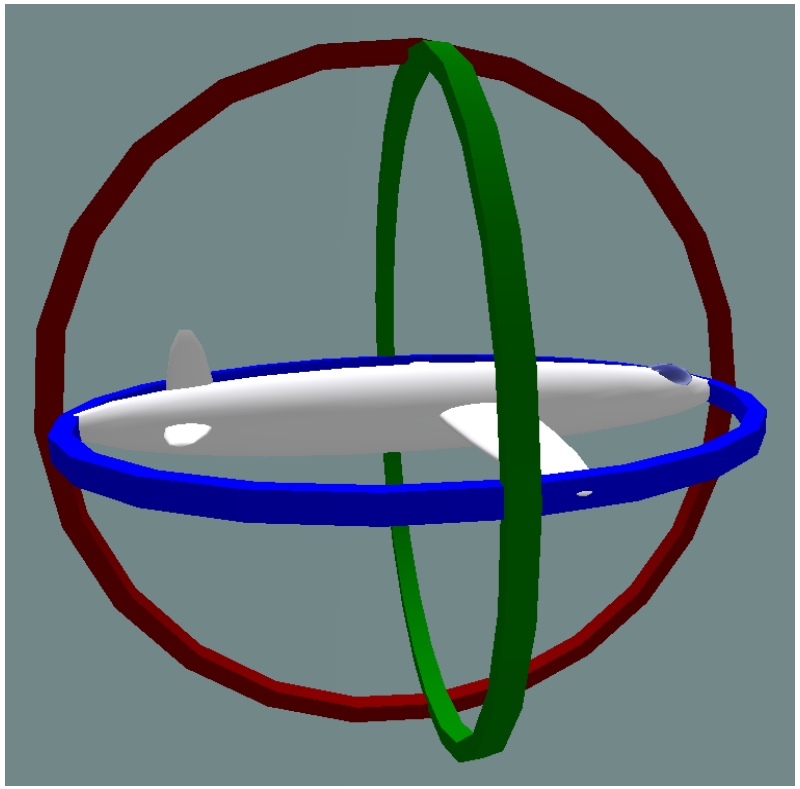
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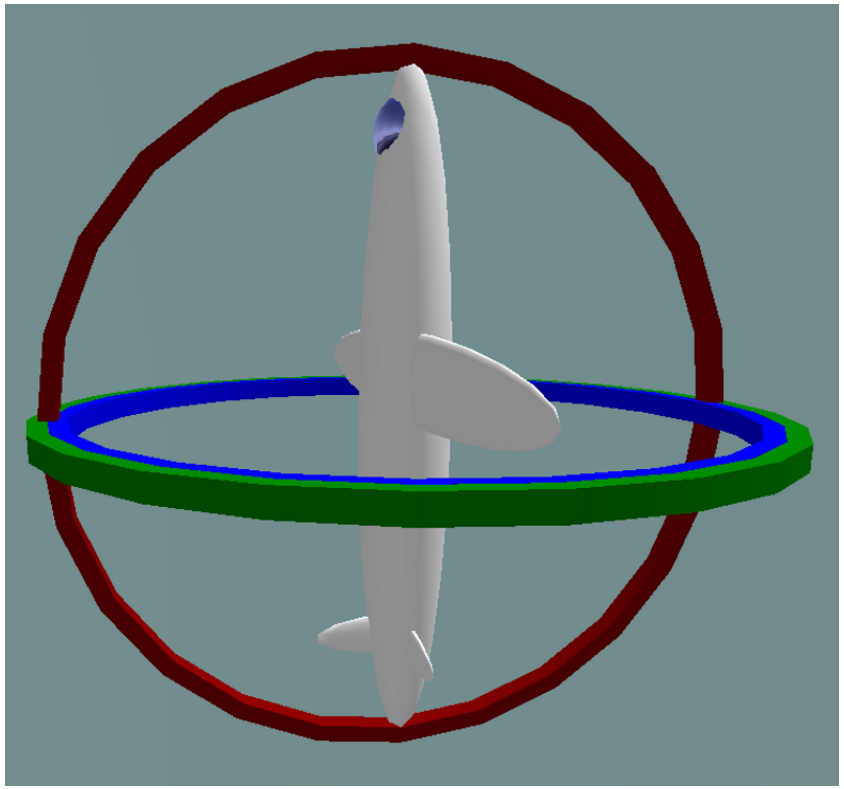
An Issue with Angle-Sets: Gimbal Lock (Example)



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Gimbal Lock





Gimbal Lock: When does it happen?

Gimbal lock can happen for **two different reasons**:

- **Mathematically**: in representations that map 3D orientation to sets of three angles (i.e. **angle-sets**).

- **Physically:**
  - In **mechanisms** that decouple orientation into several angles.
    (e.g. the airplane gunner example).
  - In **measurement** systems that decouple quantities related to orientation into several angles.
    (e.g. Inertial Measurement units → see Apollo 11)

Rotation Matrix to Fixed Angles X-Y-Z (I)

$$ {}^{A}_{B}\boldsymbol{R}_{Z'Y'X'}(\alpha,\beta,\gamma) = {}^{A}_{B}\boldsymbol{R}_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

$$ \cos\beta = \sqrt{{r_{11}}^2 + {r_{21}}^2}, $$

$$ \sin\beta = -r_{31} \longrightarrow \tan\beta = \frac{\sin\beta}{\cos\beta} \longrightarrow \beta = \operatorname{atan}\frac{-r_{31}}{\sqrt{{r_{11}}^2 + {r_{21}}^2}} $$

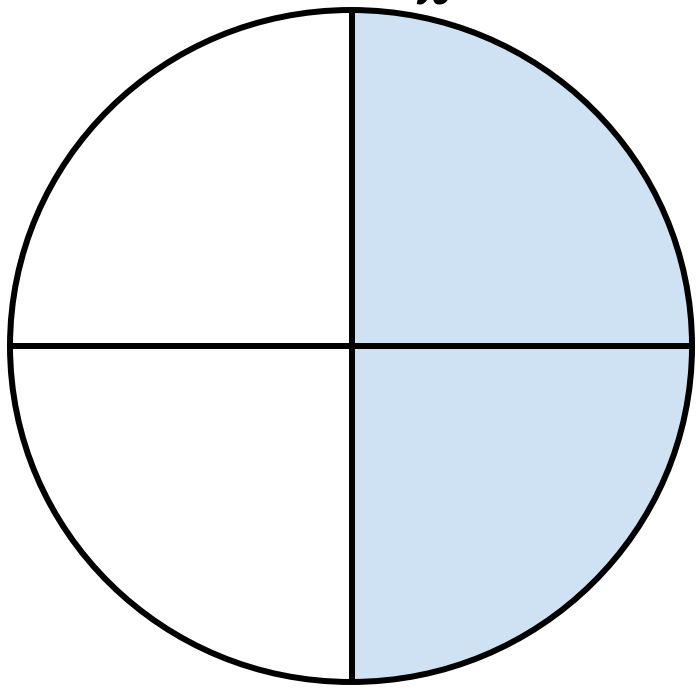
$$ \sqrt{(\cos\alpha\cos\beta)^2 + (\sin\alpha\cos\beta)^2} = \sqrt{(\cos\beta)^2\left((\cos\alpha)^2 + (\sin\alpha)^2\right)} = \cos\beta $$

$$ \beta = \operatorname{atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right) $$

atan2 is a **better version** of atan that **takes into account the full circle.**

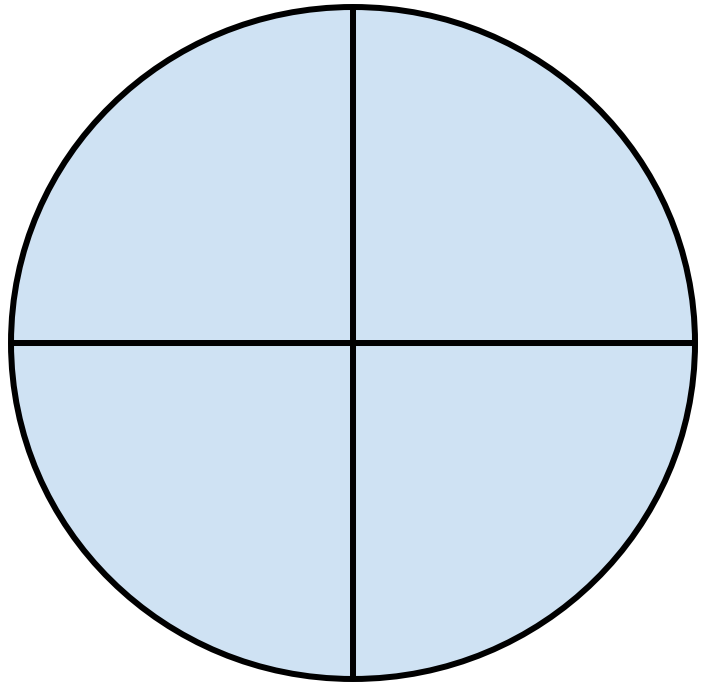
atan vs atan2

$\mathbf{atan}\left(\frac{y}{x}\right)$



Returns angles between $\frac{\pi}{2}$ and $-\frac{\pi}{2}$.

$\mathbf{atan2}(y, x)$



Returns angles in the full circle (0 to 2π).

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Rotation Matrix to Fixed Angles X-Y-Z (II)

$$ {}^{A}_{B}\boldsymbol{R}_{Z'Y'X'}(\alpha,\beta,\gamma) = {}^{A}_{B}\boldsymbol{R}_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

$$ r_{21} = \sin\alpha\cos\beta \longrightarrow \sin\alpha = \frac{r_{21}}{\cos\beta} $$

$$ r_{11} = \cos\alpha\cos\beta \longrightarrow \cos\alpha = \frac{r_{11}}{\cos\beta} $$

$$ \alpha = atan2\left(\frac{r_{21}}{\cos\beta}, \frac{r_{11}}{\cos\beta}\right) $$

$$ r_{32} = \cos\beta\sin\gamma \longrightarrow \sin\gamma = \frac{r_{32}}{\cos\beta} $$

$$ r_{33} = \cos\beta\cos\gamma \longrightarrow \cos\gamma = \frac{r_{33}}{\cos\beta} $$

$$ \gamma = atan2\left(\frac{r_{32}}{\cos\beta}, \frac{r_{33}}{\cos\beta}\right) $$

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Rotation Matrix to Other Angle-Sets

We have seen how to convert a **rotation matrix** to **fixed angle X-Y-Z**.

How do we do it for **other kinds of angle sets**?

- We can **use Fixed angle to Euler angle equivalences**.
  - e.g. Fixed angle X-Y-Z = Euler angle Z-Y-X

- We can use **similar logic** to the previous slides **for other angle combinations**:
  1. **Isolate sine** and **cosine** for one angle.
  2. **Use atan2** to compute the angle.
  3. **Repeat** for other angles.

Part III: Equivalent Angle-Axis

Equivalent Angle-Axis (Euler Vector)

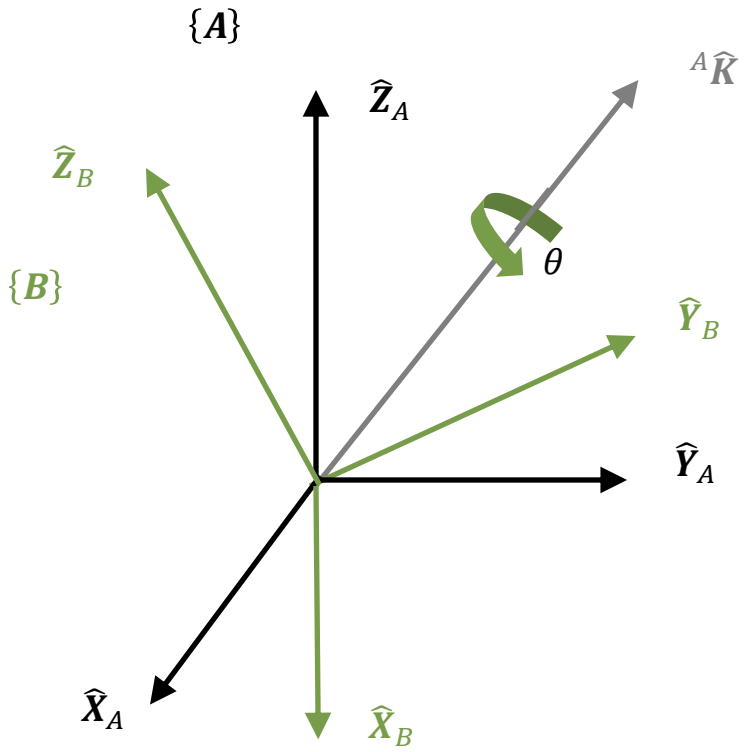
Combination of:

- **Axis**: **unit vector** that represents the **direction around which we rotate**.
- **Angle**: **scalar** that represents **how much** we rotate, using the **right-hand rule**.

Can be **given as**:

$$\theta,\ \hat{\boldsymbol{K}} = \begin{bmatrix} k_x \\ k_y \\ k_z \end{bmatrix} \quad \text{or} \quad \boldsymbol{K} = \theta\hat{\boldsymbol{K}} = \begin{bmatrix} \theta k_x \\ \theta k_y \\ \theta k_z \end{bmatrix}$$

separate **combined***



**Universal Robots uses this representation in their Graphical User Interface*

Equivalent Angle-Axis to Rotation Matrix

When we defined $\boldsymbol{R}_X(\theta),\ \boldsymbol{R}_Y(\theta),\ \boldsymbol{R}_Z(\theta)$ in Lecture 4, we were already doing this.

We can do this for **any arbitrary angle:**

$$
{}^{A}_{B}\boldsymbol{R}_K(\theta) = \begin{bmatrix} k_x k_x v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y k_y v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z k_z v\theta + c\theta \end{bmatrix}, \text{ with } v\theta = 1 - c\theta
$$

Deriving this is **quite tedious**. You can try this as an exercise (2.6).

Rotation Matrix to Equivalent Angle-Axis

Given a rotation matrix:

$$ {}^{A}_{B}\boldsymbol{R}_{K}(\theta) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

An angle-axis pair can be computed as:

$$ \theta = \operatorname{acos}\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right) $$

$$ \widehat{\boldsymbol{K}} = \frac{1}{2\sin\theta}\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} $$

Note that:

$$ \left(\theta, \widehat{\boldsymbol{K}}\right) = \left(-\theta, -\widehat{\boldsymbol{K}}\right) $$

Recap: What have we discussed today

- The **problems** with **mathematically modeling** 3D **orientations/rotations**.
- **Angle-set conventions**
  - **Fixed angles**
  - **Euler angles**
  - **Gimbal Lock**
- **Equivalent Angle-Axis**
- **Conversions** between representations

Take home message:

There are **various ways to represent orientation** in 3D space.

When **choosing** one or another, it is important to keep in mind the **application** (e.g. calculations or visualization) and the **limitations of the representation** (e.g. **Gimbal lock**).

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Thank you for today.

Iñigo Iturrate

Ø27-604-3

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